Do EIA Gasoline-Demand and Inventory Surprises Predict Energy Futures Returns?

A point-in-time event study and pre-registered out-of-sample trading test for WTI crude oil (CL) and RBOB gasoline (RB) futures, 1996–2024

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Abstract. Every Wednesday the U.S. Energy Information Administration (EIA) publishes weekly gasoline demand and petroleum inventories. I construct model-based surprises in these data using only information available before each release, including *first-published* values recovered from 793 archived reports, and test whether they move NYMEX WTI and RBOB futures on release day and whether they predict returns after the release becomes public. Release windows are built from documented release schedules, returns are computed within the futures contract actually held, and the out-of-sample period (January 2017 to April 2024) was evaluated once, after the full specification was committed and tagged in git. Gasoline-inventory surprises move RBOB futures on release day out of sample (−0.74% per standard deviation, t = −3.67, Holm-adjusted p = 0.004 across 16 pre-registered tests). There is no evidence of post-release drift: all eight drift coefficients have Holm p = 1.00, and walk-forward forecasts underperform the historical mean. The pre-registered strategy, which earned a net Sharpe ratio of +0.35 in development validation, earned −0.50 out of sample (95% bootstrap CI [−1.14, +0.14], Deflated Sharpe Ratio 0.0008, Hansen SPA p = 1.00). Weekly EIA information appears to be incorporated by the first settlement after the release.

Keywords: event study; commodity futures; point-in-time data; look-ahead bias; Newey–West; multiple testing; walk-forward forecasting; Deflated Sharpe Ratio; SPA test; pre-registration.

Table 1. Summary of findings. Q1 is a diagnostic of the surprise measure; Q2 and the strategy are the tradable hypotheses.

Question	Locked-test result (2017-01 to 2024-04)	Verdict
Q1: release-day reaction	RB ← gasoline-stock surprise: −0.74% per sd, t = −3.67, Holm p = 0.004	Significant; replicates development
Q2: post-release drift	All 8 drift coefficients Holm p = 1.00; OOS R^2 −1.5% (CL), −1.8% (RB)	No evidence
Trading strategy (net)	Sharpe −0.50, CI [−1.14, +0.14], DSR 0.0008, SPA p = 1.00	Fails all pre-registered criteria

All numbers are produced by the project's code and stored in reports/tables/; reproduction commands are in Appendix A.

1. Research question and economic motivation

The EIA's *Weekly Petroleum Status Report* (WPSR) is among the most closely watched scheduled releases in commodity markets. Weekly *gasoline product supplied* is the best high-frequency public proxy for U.S. transportation fuel demand, and inventories are the buffer between supply and demand. Under the *theory of storage*, an unexpected inventory build signals looser balances and lowers the convenience yield of physical barrels, so prompt futures prices should fall; the effect should be larger when stocks are scarce.

Predictable variation (seasonality, holidays, trend) is already in the futures curve, so only the unexpected component of each report can move prices. The study separates two questions that are often conflated:

- **Q1 – announcement reaction (diagnostic).** Does the return from the last settlement before the release to the first settlement after it respond to the surprise? This is not tradable with daily data, but if it fails the surprise measure is too noisy for Q2 to be informative.
- **Q2 – post-release drift (tradable).** From the first settlement after the release to the last settlement before the next release, does the price keep moving (under-reaction) or revert (over-reaction)? Theory does not fix the sign, so tests are two-sided.

Announcement effects of EIA inventory news are documented (e.g. Halova, Kurov and Kucher, 2014). The tradable post-release question under realistic execution timing is studied far less, and it is where look-ahead bias is easiest to introduce.

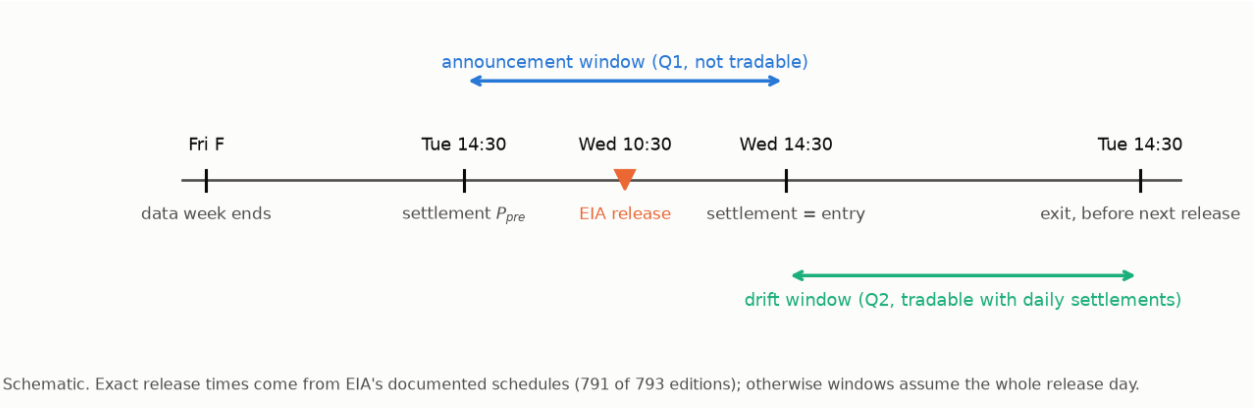


Figure 1. Event windows for a standard week. The announcement window brackets the release; the drift window starts at the first settlement after the release and ends before the next release, so a holding period never contains two announcements.

2. Data sources and point-in-time construction

Table 2. Data. All 1,105 source files are logged with URL, download time and SHA-256 hash. No vendor data was used.

Data	Source	Coverage
Weekly fundamentals: gasoline product supplied; crude (ex-SPR), gasoline and distillate stocks	EIA bulk history files	1982/1991 – 2026-10
NYMEX settlements, CL and RB contracts 1–3	EIA (discontinued after 2024-04-05)	1983/2005 – 2024-04-05
Release dates of 793 WPSR editions; first-published Table 1 values	EIA WPSR archive	2011-08 – 2026-10
Release-time schedules (live page + 194 archived copies)	EIA schedule page via the Internet Archive	2006 – 2026
Contract expiries (96 calendar copies) and contract rules	CME product calendars; NYMEX Rulebook ch. 200 and 191	1983 – 2024

Release times are documented, not assumed. A time is attached to a release only if an EIA schedule in force for that period states it *and* the observed release date matches the scheduled date; 791 of 793 editions qualify. For the remaining two the release is treated as an interval covering the whole day: windows *close* at the earliest possible release time and *open* after the latest, so uncertainty can only delay entry. Holiday release times changed across years (10:30, 11:00, 12:00, 17:00), so none is assumed. For years before the archive (pre-August 2011) a conservative rule assumes each report was public by Friday F+7 at 14:29; on all observed releases since 2011 it held in 635 of 635 non-holiday weeks (holiday weeks are excluded by construction).

First-published values, not revisions. Comparing the archived first prints with today's history file revealed that EIA restated commercial crude stocks in October 2016 (the next edition re-printed the prior week 30.6 million barrels lower) and applied the new definition backwards: crude levels match in only 10 of 283 development weeks, and the weekly *change* the market saw differs in 61 of 283. Surprises therefore use each edition's own printed values wherever an archived edition exists.

Contract-aware returns. EIA's 'contract 1' silently rolls at expiry. Returns and P&L are computed within the contract actually held, rolling five business days before the exchange-documented expiry, in dollars per contract (WTI settled at −$37.63 on 2020-04-20). Expiries come from CME-published last-trade dates where archived, and otherwise from the rulebook rule on a scheduled exchange calendar; the rule matches CME on 68/68 RB and 119/123 CL contracts where both exist.

3. Methodology

3.1 Surprises

For each weekly series, y_t is the change in stocks or the level of demand, and the surprise is the forecast error of an expectation built from data published before release t:

$$s_t = y_t - \mathbb{E}\left[\, y_t \mid y_{t-1}, y_{t-2}, \ldots \right], \qquad z_t = s_t / \mathrm{sd}(s_{<t}) \tag{1}$$

Three pre-registered expectation models are compared: the five-year same-week mean (EIA's own benchmark); a seasonal regression with trend, three Fourier harmonics and holiday dummies on a trailing five-year window; and the same regression plus an AR(1) on its residuals. Standardising by the expanding standard deviation of *past* surprises avoids leaking the full-sample scale.

3.2 Hypothesis tests

$$r^{w}_{m,t} = \alpha + \sum_{j=1}^{4} \beta_j z_{j,t} + \delta' x_{m,t} + \varepsilon_t \tag{2}$$

for market m ∈ {CL, RB} and window w ∈ {announcement, drift}. The controls $x_{m,t}$ (drift only) are one- and four-week momentum, 20-day realised volatility and the front-to-second contract carry, all measured at entry. Standard errors are Newey–West with lag $L = \lfloor 4(n/100)^{2/9} \rfloor$, robust to heteroskedasticity and serial correlation. The 16 coefficients (2 markets × 2 windows × 4 surprises) form one pre-registered family, corrected with Holm (family-wise error) and Benjamini–Hochberg (false discovery rate).

3.3 Forecast evaluation

Drift returns are forecast walk-forward: models are refit every January on events whose *exit* settlement precedes the refit date. Models are M0 (historical mean), M1 (price controls), M2 (M1 plus surprises), M3 (ridge on M2's regressors), M4 (logistic, direction) and M5 (gradient boosting, run only if a pre-registered trigger fired). Accuracy is measured by the out-of-sample R^2 of Campbell and Thompson and, for nested models, the Clark–West statistic:

$$R^2_{OOS} = 1 - \mathrm{MSE}(\hat{y}_t) \,/\, \mathrm{MSE}(\bar{y}_{<t}) \tag{3}$$

$$f_t = (y_t - \hat{y}_{1,t})^2 - \left[(y_t - \hat{y}_{2,t})^2 - (\hat{y}_{1,t} - \hat{y}_{2,t})^2 \right], \qquad \mathrm{CW} = \bar{f} / \widehat{\mathrm{se}}_{\mathrm{HAC}}(\bar{f}) \tag{4}$$

where model 1 is nested in model 2; a positive Clark–West statistic indicates that the larger model adds predictive information after correcting for its estimation noise.

3.4 Strategy inference

Sharpe-ratio uncertainty is measured with the stationary bootstrap (Politis and Romano, 1994; mean block length eight weeks). Selection across specifications is addressed with the Deflated Sharpe Ratio (Bailey and López de Prado, 2014), which compares the observed per-period Sharpe ratio with the expected maximum of N trials under the null, and with Hansen's (2005) SPA test across all tradable variants. Year-block permutations and turnover-matched random signs provide additional null distributions.

$$\mathrm{DSR} = \Phi\left((\widehat{SR} - SR_0)\sqrt{n-1} \,/\, \hat{\sigma}_{SR} \right), \qquad \hat{\sigma}^2_{SR} = 1 - \hat{\gamma}_3 \widehat{SR} + \frac{\hat{\gamma}_4 - 1}{4} \widehat{SR}^2 \tag{5}$$

where SR_0 is the expected maximum Sharpe ratio across N independent trials under the null, and γ_3, γ_4 are the skewness and kurtosis of returns.

3.5 Research protocol

The analysis protocol was committed before any futures return was examined. Data dated 2017 or later is blocked in code until a spec-freeze git tag exists, and the test script refuses to run if the code differs from the tag. One amendment was made during development: the trading signal is the forecast's deviation from the unconditional mean, because the original scaling turned a heavily shrunk model into a constant short position. It is disclosed, and its trials are counted in the Deflated Sharpe Ratio (N = 20).

4. Exploratory analysis (development sample, 1996–2016)

All series are strongly seasonal (one-way ANOVA by week of year, $p < 0.001$). Gasoline demand peaks in late June; Labor Day weeks show demand 128 kb/d below a normal week of the same season ($t = -3.03$). Weekly changes are fat-tailed and autocorrelated; inventory changes are stationary while the crude stock level has a unit root (ADF $p = 0.85$), which is why surprises are defined on changes. The theory of storage is visible in the data: when crude stocks are high relative to their five-year norm, the curve is in contango (Spearman $\rho = -0.72$ for CL, -0.30 for RB).

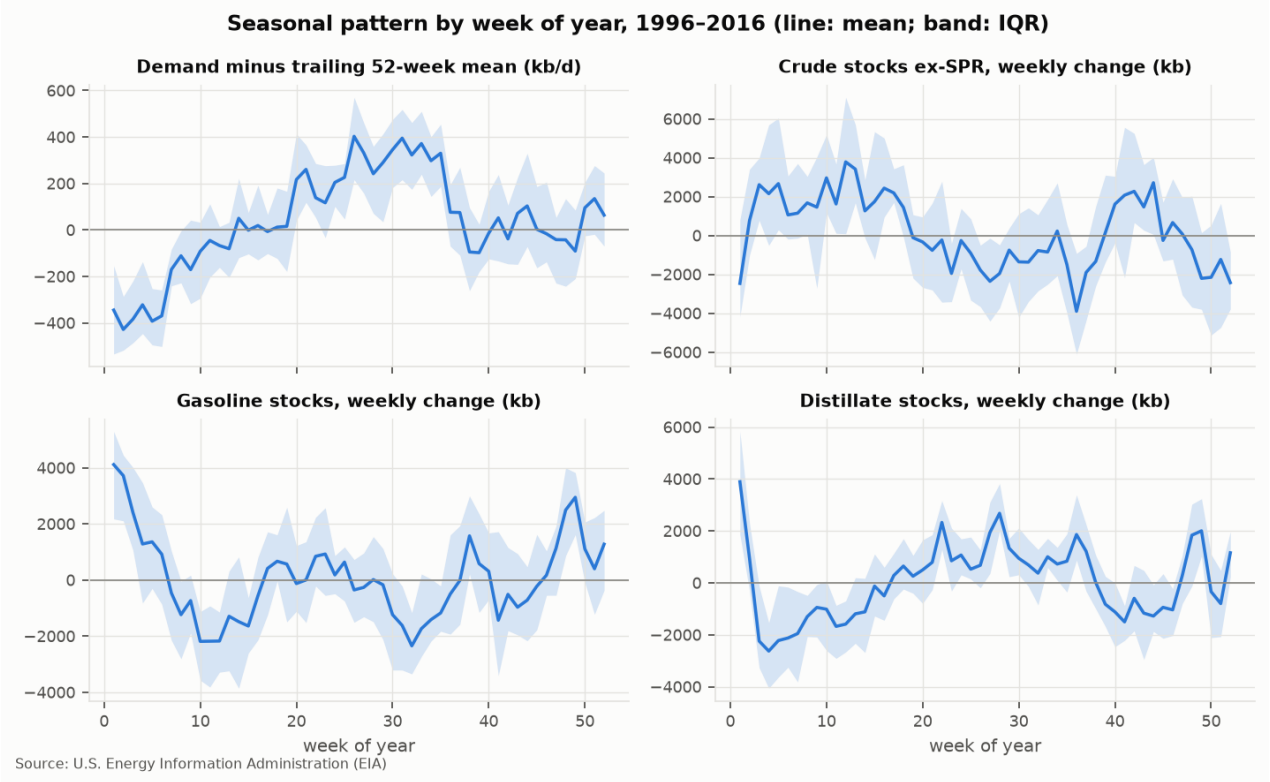


Figure 2. Seasonal pattern of demand and weekly inventory changes by week of year, 1996–2016 (line: mean; band: interquartile range).

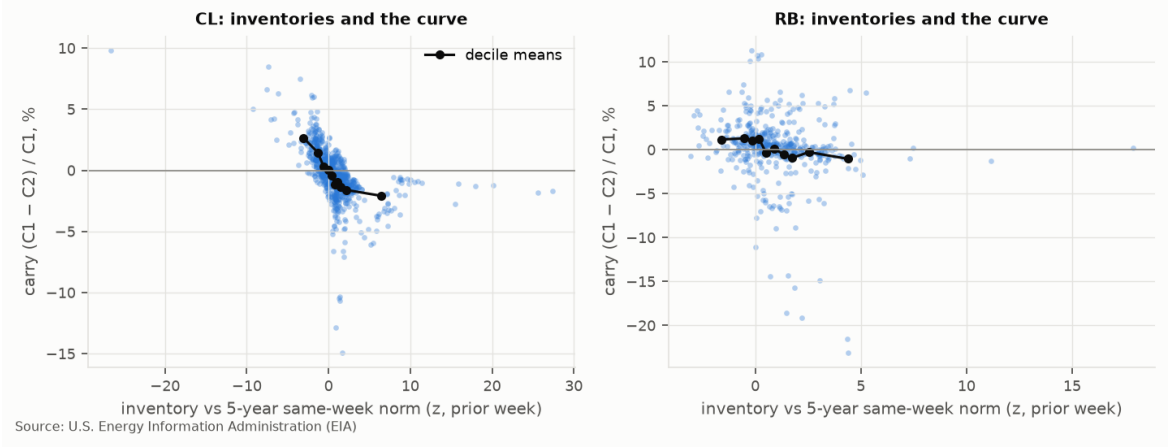


Figure 3. Theory of storage: front-to-second-contract carry against the prior week's inventory deviation from its five-year same-week norm (development data).

5. Results

5.1 Surprise models

The seasonal regression with AR(1) residuals has the lowest out-of-sample error for all four series and is selected (Figure 5). The gain is largest for demand, where RMSE falls from 410 to 232 kb/d (44%), because the five-year mean ignores the demand trend. Better expectations reduce the attenuation bias that measurement error imposes on regression slopes.

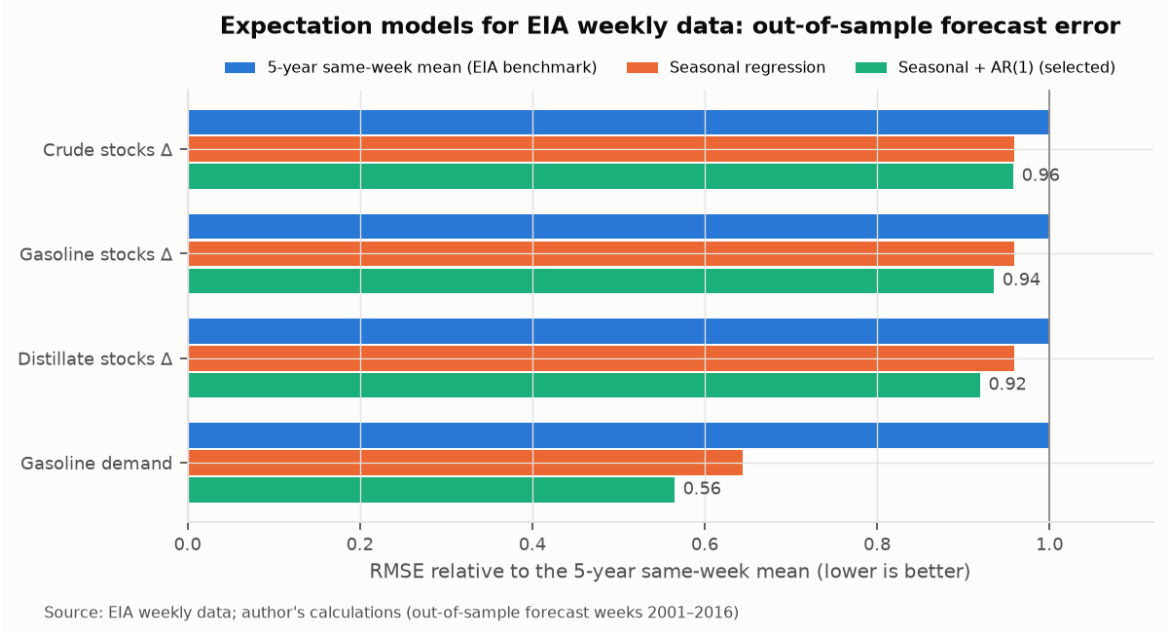


Figure 4. Out-of-sample RMSE of the three pre-registered expectation models relative to the five-year same-week mean (forecast weeks 2001–2016; fundamentals only).

5.2 Release-day reaction and post-release drift

Table 3. Selected coefficients of equation (2), percent return per one-standard-deviation surprise. Development: 2011-08 to 2016 (in-sample). Test: 2017-01 to 2024-04-05. Newey–West t-statistics; Holm adjustment across all 16 test-period coefficients.

Market, window ← surprise	Dev. coef	Dev. t	Test coef	Test t	Test Holm p
RB, announcement ← gasoline stocks	**−0.46%**	**−3.10**	**−0.74%**	**−3.67**	**0.004**
RB, announcement ← gasoline demand	0.40%	2.98	−0.27%	−1.73	1.000
CL, announcement ← crude stocks	−0.72%	−4.68	−0.17%	−1.51	1.000
CL, announcement ← gasoline stocks	0.10%	0.53	−0.61%	−2.35	0.265
CL, announcement ← gasoline demand	0.31%	2.22	−0.40%	−2.47	0.204
All 8 drift coefficients (CL and RB)	–	–	–	$\lvert t\rvert \le 1.45$	1.00

Only one of the 16 test-period coefficients survives the Holm correction: gasoline-inventory surprises lower RBOB on release day (−0.74% per standard deviation, t = −3.67), replicating the development sign; the result holds under conservative timing (t = −3.78, Holm p = 0.003). The crude → WTI reaction, strong in development (t = −4.68), shrank to t = −1.51, and the demand → RBOB coefficient changed sign (t = +2.98 in development, −1.73 in the test). Candidate explanations—information released earlier by the private API survey, the post-2015 export era, and measurement error in model-based surprises—are hypotheses, not tested results. No drift coefficient is significant, and the secondary calendar-spread and inventory-state interaction tests are not significant (all p ≥ 0.34 for the spread family).

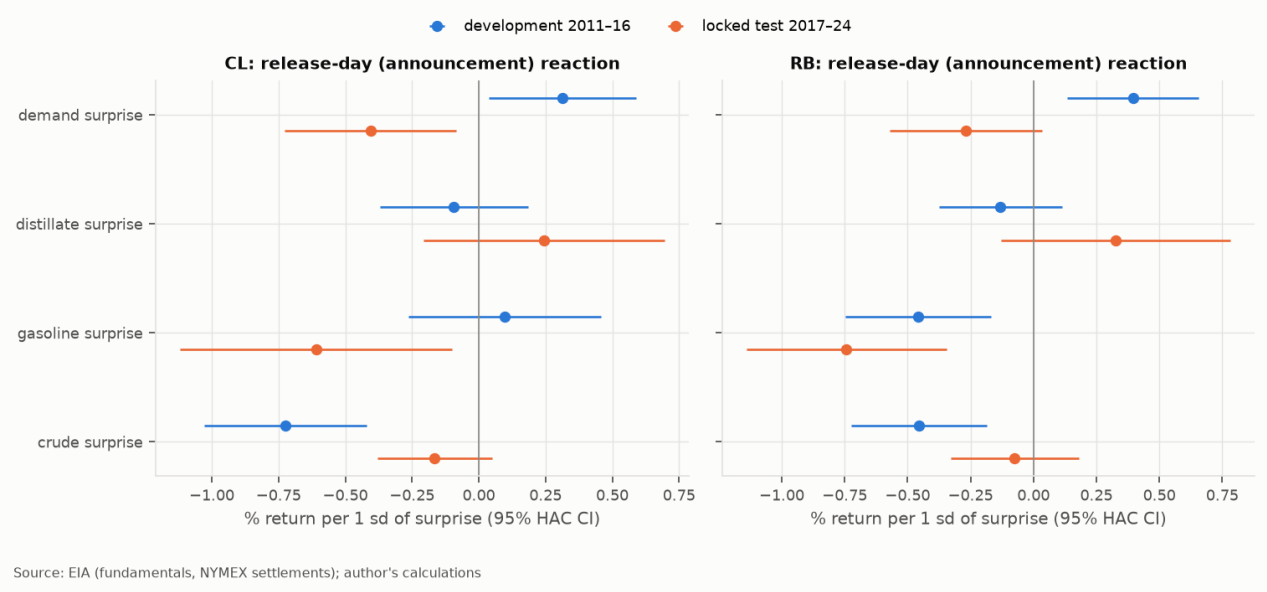


Figure 5. Release-day reaction per one-standard-deviation surprise with 95% Newey–West confidence intervals: development (2011–2016) versus the locked test (2017–2024).

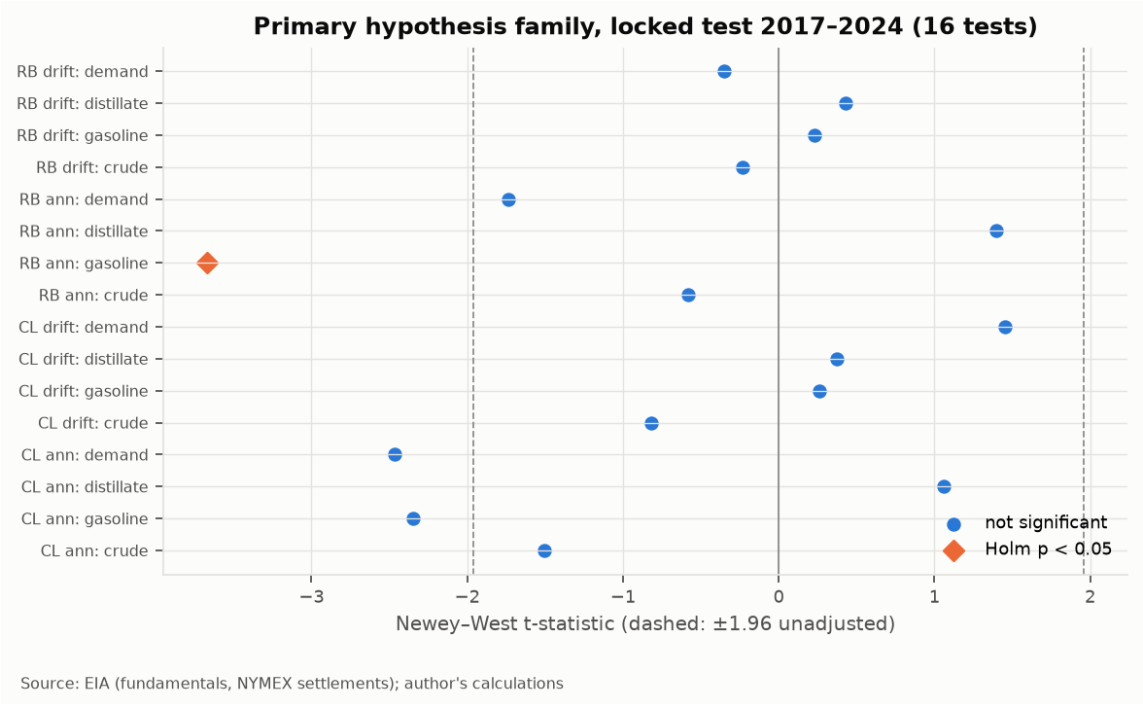


Figure 6. All 16 pre-registered test-period coefficients. The diamond marks the only effect significant after the Holm correction.

5.3 Forecasting

Table 4. Out-of-sample R^2 of drift forecasts relative to the historical mean (M0). Development: walk-forward validation 2006–2016.

Model	Dev. CL	Dev. RB	Test CL	Test RB
M1 price-only (OLS)	+0.04%	−7.03%	−3.58%	−4.31%
M2 price + surprises (OLS)	+0.40%	−8.46%	−4.01%	−6.70%
M3 ridge (selected)	+0.49%	−0.59%	−1.51%	−1.78%

Validation selected the heaviest ridge penalty on the grid (λ = 1000): out-of-sample error fell monotonically toward the unconditional mean. Gradient boosting, triggered by a nominal Clark–West result as pre-registered, did worse still in development. In the test every model underperforms the historical mean, and the direction model's hit rate (50.1% CL, 51.9% RB) is below the share of up-weeks (56.4%, 57.6%).

6. Trading strategy and backtesting methodology

The signal is the standardised *conditional* drift forecast from M3, its deviation from the training-sample mean forecast divided by the dispersion of fitted values, both from the same annual refit:

$$\hat{z}_t = \frac{\hat{r}_t - \bar{\hat{r}}_{\text{train}}}{\text{sd}(\hat{r}_{\text{train}})}, \qquad q_t = \text{sign}(\hat{z}_t)\,\mathbf{1}\{|\hat{z}_t| > k\}\;\frac{\sigma^*}{\hat{\sigma}_d\sqrt{h}\,P_t M} \tag{6}$$

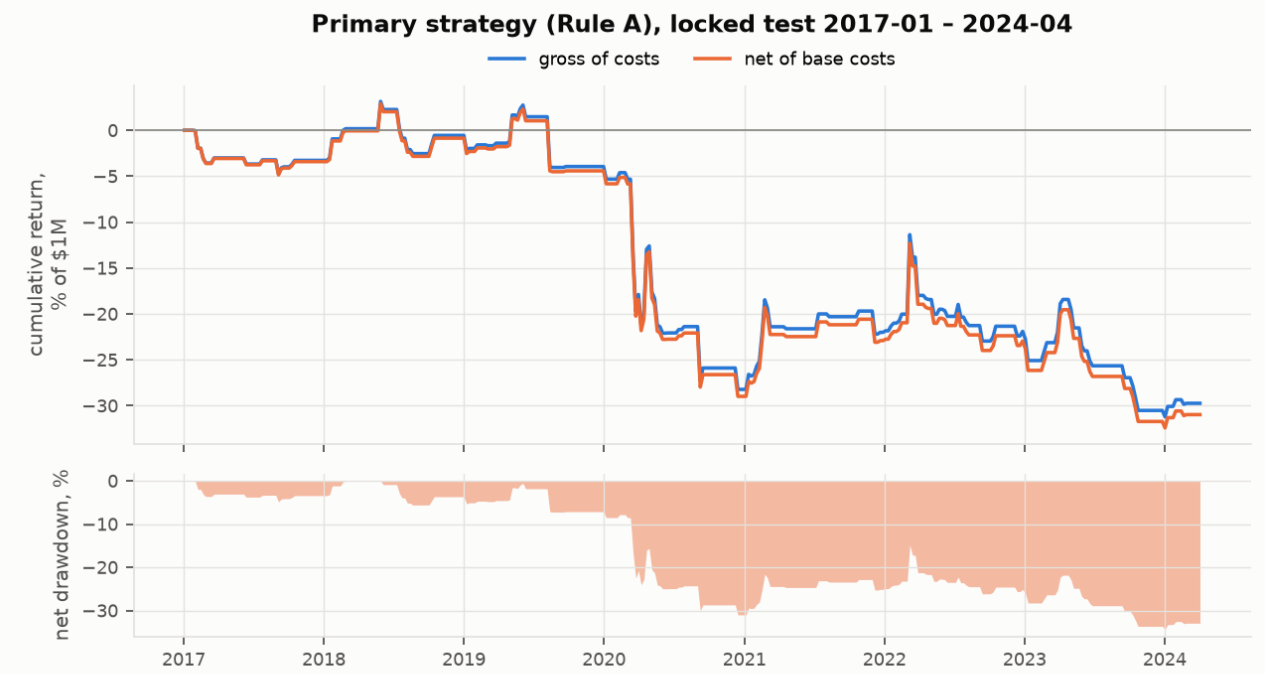
The primary rule (Rule A) is long, short or flat with threshold k = 1.5, chosen in development from {0.5, 1.0, 1.5}. Position size targets 10% annual volatility on a $1 million capital base: σ* is the weekly dollar-risk target, σ_d the trailing 60-day daily volatility known at entry, h the holding period in days, P the price and M the contract multiplier (1,000 bbl for CL, 42,000 gal for RB); contracts are rounded and gross notional is capped at twice capital. Rule B sizes positions in proportion to clip(ẑ, −2, 2), equivalent to sizing by forecast over variance. Positions are opened at the first settlement after the release and closed at the last settlement before the next one. Daily P&L is $q_{d-1}(P_h(d) - P_h(d-1))M$ in the contract h held, so rolls never book a spread between contracts; costs are charged per contract per side, including both legs of every roll. Benchmarks are always-long, a price-only strategy (Rule A on M1 forecasts) and turnover-matched random signs. Returns are weekly P&L on the fixed capital base, annualised with √52.

7. Out-of-sample performance

Table 5. Locked test, 379 weeks (2017-01 to 2024-04-05), base costs, non-compounded returns on a $1 million base.

Strategy	Gross Sharpe	Net Sharpe	Net return / yr	Volatility / yr	Sortino	Max drawdown	Costs / yr
Rule A (primary)	**−0.48**	**−0.50**	**−4.2%**	**8.4%**	**−0.65**	**−34.3%**	**$1,693**
Rule B	−0.57	−0.62	−11.7%	18.9%	−0.81	−93.6%	$8,968
Always long	0.13	0.07	1.5%	20.9%	0.09	−59.6%	$11,486
Price-only benchmark	−0.32	−0.33	−2.4%	7.3%	−0.44	−22.6%	$790

The primary strategy made 132 trades (62 CL, 70 RB) with a 43.9% win rate and lost $310k over the test; 2020 accounts for $246k, but excluding the first half of 2020 the net Sharpe ratio is still −0.27. 7 events were skipped because a settlement was missing.



Source: reports/tables/phase7/primary_weekly_test.csv (non-compounded weekly returns on a $1M base)

Figure 7. Primary strategy in the locked test: cumulative returns gross and net of base costs (percent of $1 million) and net drawdown. The two lines nearly coincide: the losses come from the signal, not from costs.

Table 6. Formal inference for the primary strategy's net returns.

Statistic	Value	Pre-registered criterion	Met
Net Sharpe, 95% stationary-bootstrap CI	−0.50 [−1.14, +0.14]; P(SR ≤ 0) = 0.93	CI above 0	No
Deflated Sharpe Ratio (N = 20 trials)	0.0008 (SR_0 = 0.092 per week)	> 0.95	No
Hansen SPA, 16 tradable variants	p = 1.00	p < 0.05	No
Year-block permutation (1,000 draws)	p = 0.91	–	–
Random signs, matched turnover (1,000 draws)	primary at the 10.8th percentile	–	–

None of the three pre-registered success criteria is met. The bootstrap interval contains zero and places 93% of its mass at or below it. The Deflated Sharpe Ratio is essentially zero: with 20 development trials, pure noise would be expected to produce a best weekly Sharpe ratio of about 0.092 (0.66 annualised), and the observed value is negative. The SPA test finds no variant that beats a zero-return benchmark after accounting for the search across variants. The null distributions in Figure 9 show the same thing from a different angle: randomly signed trades with the same turnover, or signals shuffled across years, do as well as the model's signal. Figure 8 shows that the reversal from development to test applies to every strategy, including the price-only benchmark, which is the pattern expected when development performance reflects selection on noise.

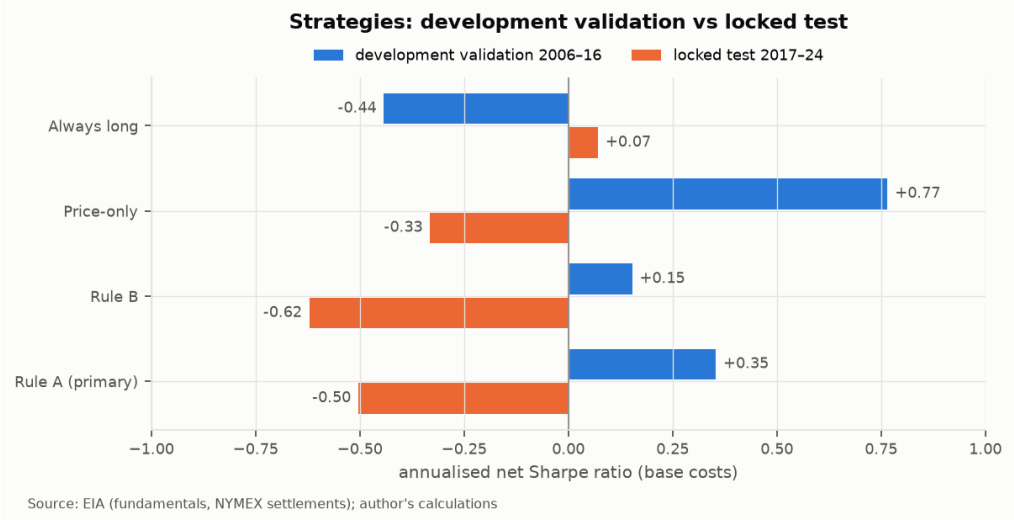


Figure 8. Net Sharpe ratios in development validation (2006–2016, walk-forward, parameters selected on this period) and in the locked test. Every positive development result reversed.

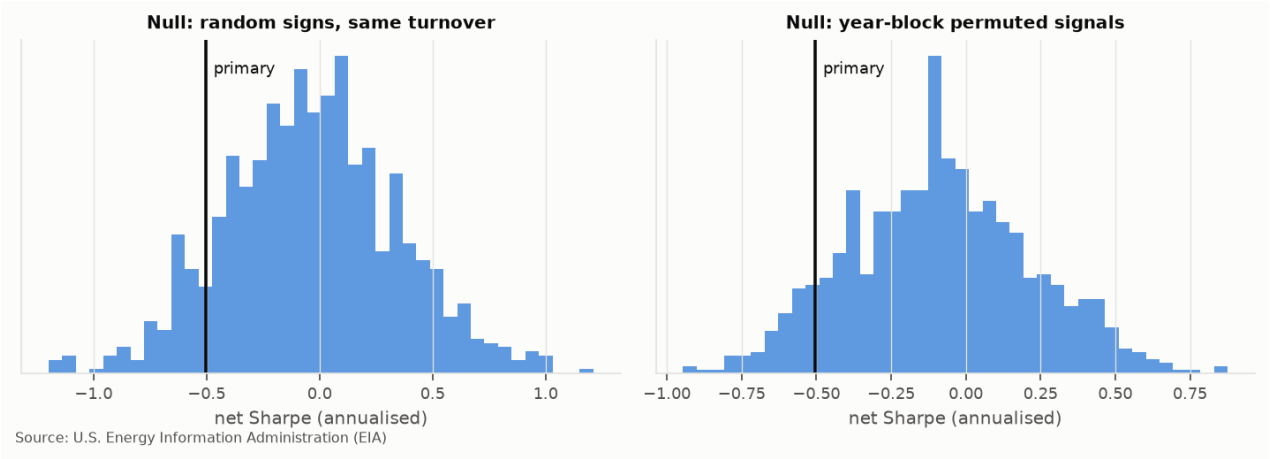


Figure 9. The primary strategy's net Sharpe against two null distributions.

8. Robustness and transaction costs

Table 7. Transaction-cost sensitivity of the primary strategy in the locked test.

Cost scenario (per contract per side)	CL	RB	Net Sharpe
Optimistic (½ tick + $1.50)	$6.50	$3.60	−0.49
Base (1 tick + $2.50)	**$12.50**	**$6.70**	**−0.50**
Conservative (2 ticks + $4.00)	$24.00	$12.40	−0.52
Stress (4 ticks + $5.00)	$45.00	$21.80	−0.55

Costs totalled $12.3k against a gross loss of $297k: the failure is in the gross signal, not in execution costs. Every tradable variant in the pre-registered grid has a negative net Sharpe ratio, from −0.61 to −0.14—across cost scenarios, thresholds, roll offsets, Thursday entry, alternative surprise models, winsorised surprises, conservative timing and sub-periods. The placebos (late Monday entry +0.14, stale signal −0.02, cross-market signal −0.22) perform no worse than the real signal, as expected if the signal carries no post-release information.

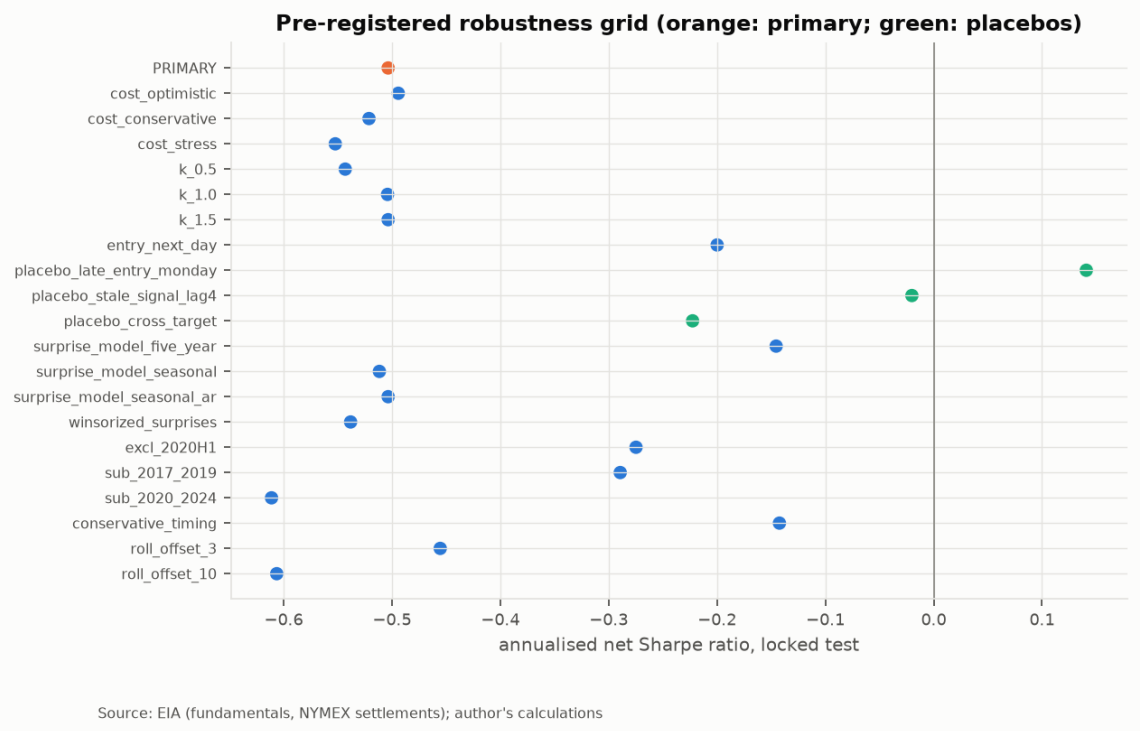


Figure 10. Pre-registered robustness grid, locked test (orange: primary specification; green: placebos).

9. Limitations

- **Daily settlements** cannot observe the minutes after the 10:30 release; any fast drift is invisible. Intraday data was deferred.
- **Power.** With 379 test weeks, a correlation of about 0.10 (≈ 2/√n) is needed for t ≈ 2, roughly a gross Sharpe ratio of 0.74; only large effects were detectable. The test ends on 2024-04-05, when EIA stopped publishing futures prices.
- **Model-based surprises** stand in for consensus forecasts, attenuating coefficients; the Tuesday API survey pre-empts part of the information.
- **Before August 2011** surprises use restated history (no first-release archive) and release times are scheduled rather than observed throughout.
- **Expiries** after 2016 (RB) and November 2022 (CL) rely on the validated rulebook rule; a possible one-day labelling ambiguity on expiry days affects roughly 2% of events by about half a percent of price.
- **One development amendment** (signal standardisation) was made after seeing development results; it is disclosed and counted in the Deflated Sharpe Ratio.

10. Conclusions

EIA weekly inventory surprises carry real release-day information: the gasoline → RBOB effect replicated out of sample under a strict multiple-testing correction. There is no evidence of a tradable post-release drift with daily settlements; every strategy variant failed the pre-registered criteria and forecasts underperformed the historical mean. Development-period success—including a price-only benchmark with a +0.77 Sharpe ratio—did not survive. Without the locked test, this project would have reported a profitable strategy. The natural next step is intraday: measuring how much of the reaction occurs in the first minutes after the release, and whether any drift exists at horizons a daily study cannot see.

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Appendix A. Reproduction

Every table and figure is produced by the repository. From the project root:

```
uv sync && uv run pytest && uv run ruff check .
uv run python -m transport_energy.data.build # data + validation
uv run python -m transport_energy.analysis.eda # Phase 4
uv run python -m transport_energy.analysis.development # Phases 5-6
TEQ_UNLOCK_TEST=spec-freeze-v1 uv run python -m transport_energy.analysis.locked_test
```

Re-running the pipeline from the frozen tag produced byte-identical result tables. The locked-test script refuses to run if the code differs from the tag.